

Energy-Efficient Self-Supervised Technique to Identify Abnormal User Over 5G Network for E-Commerce

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Abstract—Within the realm of e-commerce networks, it is frequently observed that certain users exhibit behavior patterns that differ substantially from the normative behaviors exhibited by the majority of users. The identification of these atypical individuals and the understanding of their behavioral patterns are of significant practical significance in maintaining order on e-commerce platforms. One such method for accomplishing this objective entails examining the behavioral tendencies of atypical users through the abstraction of e-commerce networks as heterogeneous information networks. These networks are then transformed into a bipartite graph that establishes associations between users and devices. The Self-Supervised Aberrant Detection Model (SAD) has been proposed within this theoretical framework as a means to identify and detect users who exhibit aberrant behavior. The SSADM methodology utilizes a self-supervised learning process that utilizes autoencoders to encode representations of user nodes. The proposed method aims to maximize a combined objective function for backpropagation while utilizing support vector data description to detect abnormalities in the representations of user nodes. In summary, many tests have been conducted utilizing both authentic network datasets and partially synthetic network datasets to demonstrate the efficacy and superiority of the SAD technique, specifically within the domain of an energy-efficient 5G network.

Index Terms—Energy efficiency, aberrant detection model, optimization technique, behavioral analysis, social networks.

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I. INTRODUCTION

WITH the continuous spread and development of the Internet, many unscrupulous merchants engage in fraudulent activities on major e-commerce platforms by manipulating a large number of users to post fake reviews, engage in malicious ordering, and other deceptive practices. These acts result in significant detriment to consumer interests. To mitigate the adverse effects caused by atypical users, the scholarly community has put forth many approaches for detecting anomalies, which have proven to be efficacious in practice.

Literature [1] proposed a density-based local outlier detection algorithm that introduces information entropy to uncover local outliers in homogeneous information networks. Literature [2] combined the K-Nearest Neighbors (KNN) outlier detection algorithm with a multi-level random forest model to detect abnormal behavior in homogeneous information networks. A person's behavior can be influenced by a variety of personal circumstances, such as age, health, illness, pain, and the effect of drugs or alcohol. Literature [3] proposed an anomaly detection algorithm based on suffix trees, which detects anomalies, based on the periodicity of anomalies in time series data. Literature [4] used autoencoders and probabilistic neural networks to achieve anomaly detection goals. Most of the current research focuses on homogeneous information networks, and many methods are still limited and not suitable for heterogeneous information networks [5]. Literature [6] proposed a dynamic anomaly detection method based on tensor representation for heterogeneous networks, which constructs a tensor index tree for classification samples and clustering, making dynamic anomaly point judgments based on whether clusters transform effectively. In the realm of literature, a study [7] has presented a method for detecting anomalies in heterogeneous information networks. The importance of different kinds of nodes is ascertained by an attention mechanism. By examining unique aggregation patterns, this makes it possible to identify abnormal users. The differentiation embeddings from heterogeneous networks are adaptively learned in the suggested technique. The literature [8] presents a method for detecting outliers in the distribution of communities within diverse information networks. Literature [9] proposed a fraud detection model called SAD for e-commerce networks, which uses autoencoders to extract behavioral patterns of fraudulent accounts in heterogeneous graphs and uses

Density-Based Spatial Clustering of Applications with Noise (DBSCAN) to detect fraud clusters. Unusual users appear as a result of several fraudulent merchants registering a large number of new users and logging in on particular devices to conduct group fraud activities. Because of the Internet's continuous expansion, many dishonest merchants utilise well-known e-commerce platforms. Literature [10] proposed a joint embedding method to capture the structure of heterogeneous graphs and detect fraud account groups in social networks. End-to-end trainable joint model for e-commerce network false review detection that combines neural random forests and autoencoders. Literature [11] proposed an end-to-end trainable joint model combining autoencoders and neural random forests to detect fake reviews in e-commerce networks. An autoencoder consists of two neural networks: a decoder that extends the latent representation back to its original size and an encoder that compresses an input, like an image, to a low-dimensional latent representation. The thousands to millions of transaction records that are generated everyday in e-commerce platforms are made up of the registration, login, purchase, review, and other behaviors of users, devices, products, and retailers. A person's Behavior can be influenced by a variety of personal circumstances, such as age, health, illness, pain, and the effect of drugs or alcohol. In light of this, we suggest the Self-Supervised Anomaly Detection Model (SAD) as a means of identifying anomalous users in e-commerce systems. The network maintains consistent detection results while also keeping the relatively complete bipartite graph structure and user behavioural aspects in user node representations by automatically supplying supervision information for anomaly detection. In order to identify fraud clusters using DBSCAN and extract fraudulent account behavior patterns from diverse graphs, an integrated embedding technique to identify fraud account clusters in social networks and capture the structure of heterogeneous graphs. In the following sections, this paper describes the behavioral characteristics of abnormal users in e-commerce networks and introduces related work, details the construction of SAD, summarizes the algorithm flow of SAD, validates and compares SAD in experiments, analyzes its time complexity, and concludes the paper while looking forward to future work. This study outlines the development of SAD, summarizes the algorithm flow of SAD, and describes the Behavioral features of atypical users in e-commerce networks. It also introduces related work. The SAD technique relies on the analysis of popularity community distribution patterns among various object kinds and employs non-negative matrix factorization for outlier detection.

This article presents the Behavioral traits of abnormal users in e-commerce networks, reviews relevant literature, goes into detail on how SAD is constructed, outlines its algorithmic flow, verifies and contrasts SAD in experiments, examines its temporal complexity, and ends the paper by looking ahead to future research. The following are the paper's most significant contributions:

1. The research abstracts the e-commerce network as a heterogeneous information network and turns it into a user-device bipartite graph in order to solve the problem of detecting aberrant users in e-commerce platforms.

2. In light of this, we propose the Supervised Anomaly Detection Model (SAD). The suggested method improves the efficiency of abnormal user detection by automatically optimizing the network in iterative cycles, while preserving the bipartite graph structure and user behavioral features.
3. Second, the usefulness and superiority of the proposed method is demonstrated through comparative experiments and analysis of SAD on three real network datasets and one semi-synthetic network dataset.

II. RELATED WORK

A. Behavioral Patterns of Abnormal Users

In e-commerce platforms, many fraudulent merchants register numerous new users and log in on specific devices to carry out collective fraudulent activities, leading to the emergence of abnormal users. Literature [7] conducted an analysis of user data on e-commerce platforms. Device clustering and activity clustering are the two main Behavioral tendencies of aberrant users. Device Clustering: Unusual users in e-commerce networks have access to a large number of computer machines for fraudulent purposes, but this Behavior has a hefty price. The majority of anomalous users usually log in on particular groupings of devices in order to control expenditures. The Supervised Anomaly Identification Model (SAD) preserves the bipartite graph structure and user behavioural data while automatically optimising the network in iterative cycles to increase the effectiveness of anomalous user identification. Device Clustering: In e-commerce networks, abnormal users control a significant number of computing devices for fraudulent activities, but this behavior comes at a high cost. To control costs, most abnormal users typically log in on specific groups of devices. Activity Clustering:

In e-commerce networks, abnormal users need to complete tasks within a short time frame, which leads to a burst of collective activity during specific time intervals. In [12], a thorough analysis of the methods for detecting novelty is provided. A detailed study of DL-based anomaly detection algorithms was recently published in [13]. A review of the data mining methods for anomaly detection is covered in [14]. Regression, rule learning, and clustering are the approaches that are covered.

B. Autoencoders

The concept of autoencoders was first introduced by Literature [15], and subsequent literature [16] provided detailed explanations. Autoencoders consist of two main parts: the encoder (E) and the decoder (D), and their structures are symmetric, meaning that the number of hidden layers in the encoder is the same as the number of hidden layers in the decoder. The encoding process of the encoder is as follows:

$$z = \sigma_e(W_1x + b_1) \quad (1)$$

The decoding process of the decoder can be described as:

$$y = \sigma_d(W_2z + b_2) \quad (2)$$

The feature representation in the hidden space is denoted by z , where $W1$ and $b1$ are the encoding weights and biases and $W2$ and $b2$ are the decoding weights and biases. The activation function is denoted by e , commonly used ones include ReLU, Sigmoid, and Tanh, while σ_d can be the same as σ_e [17].

$$J(W, b) = \Sigma(L(x, y)) = \Sigma \|y - x\|_2^2 \quad (3)$$

The encoding process maps the input signal to a feature representation in the hidden space deterministically, while the decoding process attempts to remap the feature representation in the hidden space back to the input signal [17]. Autoencoders have advantages such as a simple reconstruction process, the ability to stack multiple layers, and support from neuroscience [17]. Today, methods based on autoencoders, such as image classification [18], [19], anomaly detection [20], [21], and pattern recognition [22], are widely applied in various research fields and have achieved success. Large amount of data to obtain sufficient training samples, which is very labor and resource intensive [23]. An information entropy-based density-based technique for local outlier detection in homogeneous information networks. A multi-level random forest model combined with the KNN outlier detection technique is used to identify anomalous activity in homogeneous information networks.

Therefore, unsupervised anomaly detection has received widespread attention, and autoencoder-based anomaly detection methods are widely used in fields such as e-commerce fraud detection [24] and social network abnormal user detection [25] due to their excellent performance.

III. ANOMALY USER DETECTION METHOD FOR E-COMMERCE NETWORKS

This section introduces an anomaly user detection method for e-commerce networks - the Supervised Anomaly Detection Model (SAD). It provides a detailed description of SAD from three aspects: basic definition, model structure, and optimization stage.

A. Basic Definition

In e-commerce platforms, every user leaf records when they log in, browse, make purchases, and provide reviews. As shown in Figure 1, these records cover the relationships between different entities, such as users, devices, products, and merchants, including the behavioral traces of abnormal users. The links between various entities, including users, devices, items, and merchants, as well as the Behavioral traces of problematic users, are covered by these records, as seen in Figure 1. The e-commerce network is abstracted as a heterogeneous information network in order to more intuitively analyze the Behavioral traits of various users inside the network.

According to the first definition provided by the Heterogeneous Information Network [6], Consider a directed graph $G = (V, E; \tau, \varphi; A, R)$. In this graph, V represents the set of nodes, E represents the set of edges, τ is a function that maps each object to its object type, φ is a function that maps each relationship to its relationship type, and $\tau(v) \in$

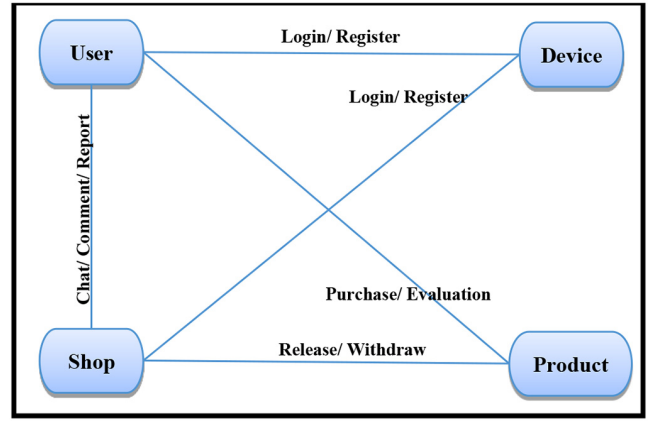


Fig. 1. Association relationships between different entities in the e-commerce network.

A indicates that each object $v \in V$ belongs to a specific object class, while $\varphi(e) \in R$ indicates that each relationship $e \in E$ belongs to a specific relationship class. An information network is referred to as a heterogeneous information network when the number of node types $|A|$ is greater than 1 or the number of edge types $|R|$ is greater than 1. Conversely, if $|A| = 1$ and $|R| = 1$, the information network is classified as a homogeneous information network.

In Section II-A, Literature [7] analyzed the login patterns of users on different devices in e-commerce platforms. We concentrate on the relationships that exist between two different types of entities: people and gadgets. End-to-end trainable joint model that blends neural random forests and autoencoders for e-commerce network false review detection. When describing different things and their relationships, heterogeneous information networks are an abstraction of the underlying real-world systems. The method aims to convert the heterogeneous information network into a bipartite graph that accurately represents the relationship between individuals and devices. This graph is going to be an essential part of our next research projects. The e-commerce infrastructure is modelled as a bipartite graph of consumers and their devices after being abstracted as a heterogeneous information network.

Definition 1 (Bipartite Graph Structure): Given a directed graph $G = (X, Y, E)$, where $X = \{x_1, x_2, \dots, x_m\}$ represents a set of m user nodes, $Y = \{y_1, y_2, \dots, y_n\}$ represents a set of n device nodes, and $E = \{e_{ij}\}_{i=1,2,\dots,m}^{j=1,2,\dots,n}$ represents a set of directed edges from X to Y . If there is an edge from x_i to y_j , then $e_{ij} = 1$; otherwise, $e_{ij} = 0$. Therefore, the bipartite graph structure can be represented as a matrix $S = [s_1, s_2, \dots, s_m]^T$, where $s_i = [e_{i1}, e_{i2}, \dots, e_{in}]$.

B. Model Structure of S-SAD

This section offers a comprehensive explanation of the model structure of S-SAD, and Figure 2 illustrates the entirety of the SAD framework.

Most users typically log in to e-commerce platforms only on their personal devices, while abnormal users tend to log in on specific groups of devices. Any Behavior that differs from what is deemed normal is termed abnormal. Psychologists utilize four broad criteria to classify conduct as abnormal:

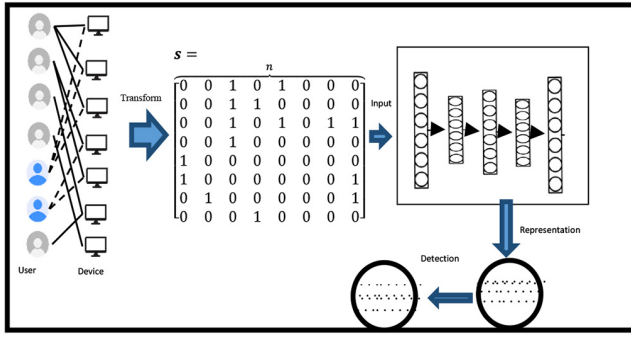


Fig. 2. The complete framework of the abnormal user detection method SAD for e-commerce networks.

statistical rarity, emotional suffering, maladaptive Behavior, and breach of social norms.

Formally, by taking the bipartite graph structure matrix S as the input to the autoencoder, the encoding process can be represented as:

$$Z = \sigma(WS + b) \quad (4)$$

Then, the user node representations Z are decoded to reconstruct the bipartite graph structure matrix $\hat{S}$, and the formal representation of the decoding process is as follows:

$$\hat{S} = \sigma(WZ + b) \quad (5)$$

where $Z = [z_1, z_2, \dots, z_m]^T$, and z_m represents the user node representation corresponding to the m -th user node in the latent space. W and b are the decoding weights and biases, and σ is the activation function. Both the encoder and decoder parts use the ReLU activation function [7]. The device similarity between user nodes is defined as:

$$\text{sim}_{-d_{ij}} = \frac{|N_i \cap N_j|}{|N_i \cup N_j|} \quad (6)$$

In which, $N_i := \{y_j \in Y : e_{ij} = 1\}$ represents the set of device nodes associated with the user node x_i .

According to activity clustering, it is known that an abnormal user group will engage in collective activities during a specific time period of the day. Therefore, the day is divided into 24 time periods, and the number of times each user logs in to different devices is counted, denoted as T for each time period. The login behavior of each user is described as $t_i = [T_0, T_1, \dots, T_{23}]$. The activity similarity between user nodes is defined as:

$$\text{sim}_{-t_{ij}} = \frac{t_i \cdot t_j}{|t_i| \times |t_j|} \quad (7)$$

Combining the two types of similarity, the behavioral similarity between user nodes in the original space is defined as follows:

$$\text{sim}_{ij} = \text{sim}_{-d_{ij}} \times \text{sim}_{-t_{ij}} \quad (8)$$

As user nodes x_i and x_j are represented in the latent space as z_i and z_j , the behavioral differences between user node representations can be defined using Euclidean distance:

$$\text{dis}_{ij} = \|z_i - z_j\|_2^2 \quad (9)$$

Furthermore, the Euclidean distance can be transformed into behavioral similarity between user node representations using the following mapping function:

$$\widehat{\text{sim}}_{ij} = \exp(-\text{dis}_{ij}) \quad (10)$$

where $\widehat{\text{sim}}_{ij}$ ranges from (0, 1). For user nodes x_i and x_j , when their distance is close to 0, $\widehat{\text{sim}}_{ij}$ is approximately 1, representing that x_i and x_j have small behavioral differences. When their distance is large enough, $\widehat{\text{sim}}_{ij}$ is approximately 0, indicating that x_i and x_j have significant behavioral differences. By narrowing the difference between $\widehat{\text{sim}}_{ij}$ and sim_{ij} , you can obtain node representations that capture user behavioral characteristics.

Backpropagation is used to optimise the joint objective function in order to offer SAD a self-supervised mechanism, which allows user node representations to autonomously provide supervision for anomaly detection. updating the weights and biases in the autoencoder network.

Next, the Support Vector Data Description (SVDD) is employed for anomaly detection on the user node representations Z .

First, the core c of the hypersphere in the latent space is computed:

$$c = \frac{1}{m} \sum_{i=1}^m z_i \quad (11)$$

Then, the Euclidean distances between each user node representation and the core are calculated:

$$d_i = \|z_i - c\|_2^2 \quad (12)$$

And a set of distances D is formed: $D = \{d_1, d_2, \dots, d_m\}$.

To find an appropriate hypersphere radius r , the normal distribution of the set D is discussed using the 3σ criterion. The following text defines the 3σ criterion.

If $x \sim N(\mu, \sigma^2)$, then:

$$P\{|x - \mu| < \sigma\} = 0. \quad (13)$$

$$P\{|x - \mu| < 2\sigma\} = 0.9445 \quad (14)$$

$$P\{|x - \mu| < 3\sigma\} = 0.9977 \quad (15)$$

Based on Equation (15), it is evident that the likelihood of a normally distributed variable x falling outside the range $(\mu - 3\sigma, \mu + 3\sigma)$ is below 0.003, a probability that is commonly regarded as being exceedingly low.

C. Optimization Phase

The joint objective function of the SAD model comprises three components. The initial component of the objective function pertains to the reconstruction error of the autoencoder, which quantifies the disparity between the original input S and the reconstructed output $\hat{S}$. Minimizing the reconstruction error helps preserve the bipartite graph structure more effectively. The definition of the objective function L_{rec} is as follows:

$$L_{\text{rec}} = \sum_{i=1}^m \|\hat{s}_i - s_i\|_2^2 \quad (16)$$

The second part of the objective function concerns the difference between $\widehat{\text{sim}}_{ij}$ and sim_{ij} . Minimizing this objective function helps obtain user node representations with behavioral characteristics. The objective function L_{sim} is defined as follows:

$$L_{\text{sim}} = \sum_{i=1, j=1}^m \|\widehat{\text{sim}}_{ij} - \text{sim}_{ij}\|_2^2 \quad (17)$$

In the paper [26], Deep SVDD optimizes the objective function of SVDD to place the majority of user nodes within the minimum hypersphere around the core, while anomalous user nodes lie outside the hypersphere. To constrain user node representations, the third part of the objective function is defined as in [26]:

$$L_{\text{svdd}} = \frac{1}{m} \sum_{i=1}^m \|z_i - c\|_2^2 \quad (18)$$

Finally, to simultaneously preserve the bipartite graph structure and user behavioral characteristics in user node representations and constrain them, the joint objective function is defined based on Equations (16) to (18). The definition of the objective function L is as follows:

$$L = L_{\text{rec}} + \alpha(L_{\text{sim}} + L_{\text{svdd}}) \quad (19)$$

where α is a hyperparameter. Minimizing L helps obtain user node representations with supervised information.

IV. SAD ALGORITHM FLOW

This section presents the algorithmic flow of SAD, providing theoretical guidance for subsequent experimental validation and comparative analysis.

V. EXPERIMENTAL RESULTS AND ANALYSIS

To validate the effectiveness and superiority of SAD, several existing baseline methods were selected for comparative experiments. The experimental environment utilized an Intel Core i7-7700 CPU with a clock speed of 3.60GHz, 8GB of memory, and a 64-bit Windows 10 operating system.

A. Baseline Methods

Firstly, the baseline methods used in the experiments are introduced, including three classical anomaly detection methods such as Isolation Forest, classic autoencoder-based anomaly detection methods, and recent anomaly detection methods like Deep FD [9] and Fraud NE [10].

B. Datasets

In this study, experiments were conducted using three real e-commerce network datasets obtained from Kaggle and one semi-synthetic e-commerce network dataset. The datasets are shown in Table I, and the subsequent descriptions offer comprehensive information regarding each dataset.

- (1) Real Dataset 1: The dataset encompasses user purchase records derived from a certain e-commerce platform. It comprises various attributes such as user IDs, user ages,

Algorithm 1 SAD Anomaly Detection Method for E-Commerce Networks

Input: Heterogeneous information network data

Output: Anomaly detection evaluation metrics results

Convert heterogeneous network data to a user-device bipartite graph and construct the bipartite graph structure matrix S using Definition 1.

for $i, j = 0$ to m

 Calculate behavioral similarity between user nodes using Equation (8).

end

Use matrix S as input to the autoencoder. Choose the optimizer as stochastic gradient descent and set the values for epochs, batch size, and learning rate (EPO, batch size, learning rate). for $i = 0$ to epoch

 Calculate user node representations using Equation (4) and complete forward propagation. Calculate behavioral similarity between user node representations using Equation (10).

 Optimize the joint objective function L using Equation (19) and complete backward propagation.

 Calculate the core of the hypersphere latent space using Equation (11).

end

for $i = 0$ to m

 Calculate distances between user node representations and the core using Equation (12) to form the set D .

end

Remove d_i values in D that fall outside the interval using principles from Equation (13) to (15). Select the maximum value in the remaining set as the radius r .

for $i = 0$ to m

 if $d_i > r$

 The node is considered an anomalous user node.

 else

 The node is considered a normal user node.

end

end

Return the anomaly detection evaluation metrics results.

end

user genders, login device IDs, purchase timings, and more relevant information. In this experiment, a subset of the original dataset was

- (2) Real Dataset 2: The dataset encompasses user purchase records obtained from a certain e-commerce platform, encompassing various user information such as IP addresses, email addresses, phone numbers, login device IDs, and more data.
- (3) Real Dataset 3: The information encompasses user purchase records obtained from a prominent online retailer, spanning the time frame between October 2019 and April 2020.
- (4) Semi-Synthetic Dataset: This dataset contains user purchase records from a certain e-commerce platform, including user IDs, user ages, user genders, login device IDs, and purchase times. It was created by randomly sampling from the original data and adding behavioral

TABLE I
E-COMMERCE NETWORK DATA SET

Data set	Number of user nodes	Number of device nodes	Number of sides
Real data set 1	190	283	5000
Real data set 2	44	106	5000
Real data set 3	237	276	3000
Semi-synthetic data set	61	89	4000

TABLE II
BEWILDERMENT MATRIX

The true situation	Test results	
	Positive example	Counterexample
Positive example	True Example (TP)	False Negative Example (FN)
Counterexample	False Positive (FP)	True Negative (TN)

records for some anomalous users, where the number of anomalous user's accounts for 5% of all users.

C. Evaluation Metrics

Since anomaly detection datasets are typically imbalanced, evaluating performance can be more complex. Finding data points or patterns in a dataset that substantially depart from the norm is known as anomaly detection. Five evaluation metrics are used to assess the performance of various methods. These metrics include Precision (P), Recall (R), F1-score, Area Under the Curve (AUC), and the G-mean.

The evaluation of measurements is conducted by employing a bewilderment matrix, exemplified in Table II. Similarly, "FP" designates the count of false positive samples that have been inaccurately classified, while "TN" represents the quantity of true negative samples that have been correctly classified.

(1) Precision is defined as:

$$P = \frac{TP}{TP + FP} \quad (20)$$

(2) Recall is defined as:

$$R = \frac{TP}{TP + FN} \quad (21)$$

(3) In general, there exists a trade-off between precision and recall. The F1-measure is computed by using the following formula:

$$F1 = \frac{2 \times P \times R}{P + R} \quad (22)$$

(4) The evaluation statistic known as AUC (Area Under the Curve) holds significant importance in the context of classification algorithms, since it quantifies the area

TABLE III
EVALUATION INDEX RESULTS OF DIFFERENT METHODS IN REAL DATA SET 1

Method name	Precision	Recall	F1-measure	AUC	G-mean
IF	0.9913	0.8959	0.9411	0.4498	0.8977
KNN	0.9913	0.91	0.9489	0.457	0.912
LOF	0.9912	0.9047	0.946	0.4543	0.9066
AE+IF	0.9916	0.8959	0.9412	0.4499	0.9012
AE+KNN	0.9913	0.9141	0.95	0.4536	0.9109
AE+LOF	0.9913	0.91	0.9495	0.47	0.9067
Deep FD	0.953	0.9749	0.9639	0.5259	0.9867
Fraud NE	0.9634	0.9715	0.9682	0.5524	0.9828
SAD	0.9793	0.9796	0.9789	0.7311	0.987

TABLE IV
EVALUATION INDEX RESULTS OF DIFFERENT METHODS IN REAL DATA SET 2

Method name	Precision	Recall	F1-measure	AUC	G-mean
IF	0.5845	0.6172	0.5296	0.5327	0.7538
KNN	0.5981	0.6127	0.5294	0.5317	0.7573
LOF	0.5933	0.6108	0.5275	0.5299	0.7555
AE+IF	0.5848	0.64	0.5423	0.4965	0.7702
AE+KNN	0.619	0.605	0.5348	0.533	0.7689
AE+LOF	0.6164	0.6045	0.537	0.5436	0.7632
Deep FD	0.8107	0.888	0.8411	0.5221	0.9422
Fraud NE	0.8307	0.8992	0.8623	0.5422	0.9466
SAD	0.9175	0.9104	0.9145	0.6016	0.9524

beneath the Receiver Operating Characteristic (ROC) curve. The calculation is performed as follows:

$$TPR = \frac{TP}{TP + FN} \quad (23)$$

$$FPR = \frac{FP}{FP + TN} \quad (24)$$

(5) G-mean is used for imbalanced datasets and is defined as:

$$G \text{ mean} = \sqrt{R \times TNR} \quad (25)$$

where TNR (True Negative Rate) represents the proportion of detected negatives among all actual negatives, calculated as:

$$TNR = \frac{TN}{TN + FP} \quad (26)$$

D. Experimental Validation and Comparative Analysis

The tests were carried out utilizing the TensorFlow framework, wherein the latent space dimension was configured to be 10. The experimental findings are displayed in Tables III to 6, whereby the most favorable outcomes for each statistic are highlighted in bold.

A comparative analysis is performed between SAD and baseline methods in the four experiments. Table III presents the evaluation metric results for different methods on Real Dataset 1, where SAD performs better in Recall, F1-measure, AUC, and G-mean.

TABLE V
EVALUATION INDEX RESULTS OF DIFFERENT
METHODS IN REAL DATA SET 3

Method name	Precision	Recall	F1-measure	AUC	G-mean
IF	0.9902	0.8954	0.9404	0.4498	0.8975
KNN	0.9902	0.9154	0.9514	0.4599	0.9176
LOF	0.9903	0.9007	0.9434	0.4525	0.9028
AE+IF	0.9902	0.8955	0.9405	0.4498	0.9135
AE+KNN	0.9906	0.9006	0.9587	0.4544	0.9274
AE+LOF	0.9903	0.9138	0.9463	0.4559	0.932
Deep FD	0.9581	0.9789	0.9684	0.5001	0.9831
Fraud NE	0.9488	0.9636	0.9581	0.5009	0.9812
SAD	0.9749	0.9767	0.9742	0.6487	0.984

TABLE VI
EVALUATION INDEX RESULTS OF DIFFERENT
METHODS IN SEMI-SYNTHETIC DATA SETS

Method name	Precision	Recall	F1-measure	AUC	G-mean
IF	0.8933	0.8482	0.8701	0.4473	0.871
KNN	0.8941	0.8611	0.8772	0.4542	0.8843
LOF	0.8987	0.863	0.8803	0.4825	0.8849
AE+IF	0.8937	0.8491	0.879	0.4626	0.8718
AE+KNN	0.894	0.8606	0.8792	0.4552	0.8908
AE+LOF	0.9037	0.8692	0.8792	0.5166	0.8931
Deep FD	0.9117	0.9414	0.9246	0.4971	0.9702
Fraud NE	0.9101	0.9491	0.9295	0.4992	0.9712
SAD	0.911	0.9513	0.9313	0.5218	0.9726

Table IV lists the evaluation metric results for different methods on Real Dataset 2, where SAD outperforms the baseline methods in various metrics.

Table V presents the evaluation metric results for different methods on Real Dataset 3. KNN achieves the highest Precision in this experiment, while Deep FD performs best in Recall. SAD outperforms the baseline methods in F1-measure, AUC, and G-mean, and its performance is second only to Deep FD in Recall.

Table VI displays the evaluation metric results for different methods on the semi-synthetic dataset. Deep FD performs best in Precision, with SAD closely following. SAD outperforms the baseline methods in other evaluation metrics. Improved feature representations of the data through self-supervised learning can improve computer vision model performance. Furthermore, the use of an SSL method improves a model's capacity to learn in the absence of the structure that labelled data provides.

Analyzing the experimental results, SAD demonstrates its effectiveness and superiority in various evaluation metrics across different datasets. Within this theoretical framework, the Self-Supervised Aberrant Detection Model (SAD) has been presented to recognize and identify people who display aberrant behavior. Autoencoders are used in the self-supervised learning process of the SSADM methodology to encode user node representations. the usage of the Self-Supervised Anomaly Detection Model (SAD) in e-commerce systems to detect unusual user Behavior. It finishes forward propagation

TABLE VII
ORDINAL VALUE TABLE

Method name	Real data set 1	Real data set 2	Real data set 3	Semi-synthetic dataset	Mean ordinal value
IF	9	7	9	9	8.5
KNN	6	8	5	7	6.5
LOF	7	9	7	4	6.75
AE+IF	8	4	8	8	7
AE+KNN	4	6	3	5	4.5
AE+LOF	5	5	6	6	5.5
Deep FD	3	3	2	3	2.75
Fraud NE	2	2	4	2	2.5
SAD	1	1	1	1	1

by encoding user node representations using autoencoders. This indicates that the anomaly detection method with a self-supervised learning mechanism exhibits strong detection performance and generalization capabilities.

To comprehensively compare the generalization capabilities of all methods, a Friedman test is performed on the F1-measure evaluation metric results for different methods across different datasets.

The null hypothesis (H0) and alternative hypothesis (H1) for the Friedman test are as follows:

H0: There is no difference in F1-measure evaluation metric results among different methods across different datasets, meaning that the performance of different methods is the same.

H1: There is a difference in F1-measure evaluation metric results among different methods across different datasets, indicating that the performance of different methods varies.

First, the F1-measure evaluation metric results for different methods across different datasets are ranked, and the average ranks are calculated for each row to obtain the average rank, as shown in Table VII.

Assuming the comparison of k methods on N datasets, let r_i represent the average rank of the i -th method. The r_i follows a normal distribution with a mean of $(k+1)/2$ and a variance of $(k^2 - 1)/12$. This leads to a χ^2 distribution:

$$\tau_{\chi^2} = \frac{12N}{k(k+1)} \left(\sum_{i=1}^k r_i^2 - \frac{k(k+1)^2}{4} \right) \quad (27)$$

Based on Equation (27), the F-distribution can be derived:

$$\tau_F = \frac{(N-1)\tau_{\chi^2}}{N(k-1) - \tau_{\chi^2}} \quad (28)$$

where τ_F follows an F-distribution with degrees of freedom $k-1$ and $(k-1)(N-1)$ [27]. At a significance level $\alpha = 0.05$ and degrees of freedom of 8, by calculation, $\tau_F = 13.94$, which is greater than the critical value 2.355 [27]. This confirms that the null hypothesis H0 is not valid, indicating that there is a difference in the performance of different methods.

Next, a sensitivity analysis of the hyperparameter α is performed. When α takes different values in the range (0,1), the evaluation metric results of SAD on different datasets are observed to see if they change. The statistical distribution

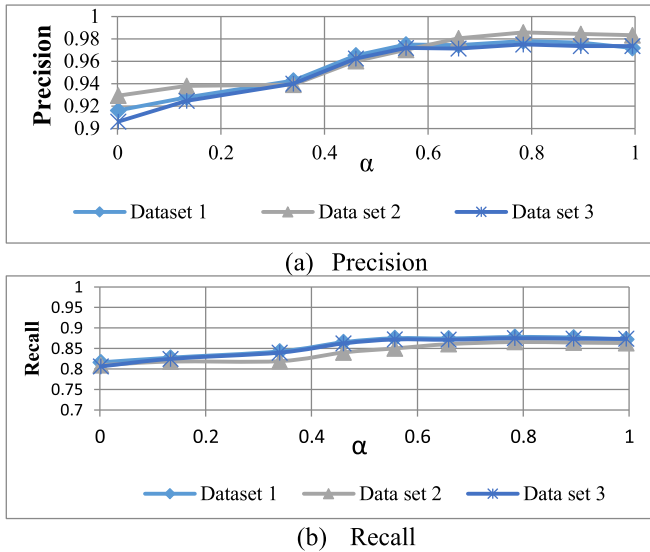


Fig. 3. Different evaluation index results of SAD when α changes in data set 1-3.

TABLE VIII
FIGMEASURE EVALUATION INDEX RESULTS OF SAD WHEN THE ACTIVATION FUNCTION CHANGES IN REAL DATA SET 1

Activation Function	Precision	Recall	F1-Measure	AUC	G-Mean
Relu	0.9916	0.9796	0.9789	0.7311	0.987
Sigmoid	0.9796	0.9809	0.9779	0.7537	0.9867
Tanh	0.9777	0.9785	0.9781	0.7072	0.9867

TABLE IX
F1-MEASURE EVALUATION INDEX RESULTS OF SAD WHEN THE ACTIVATION FUNCTION CHANGES IN REAL DATA SET 2

Activation function	Precision	Recall	F1-measure	AUC	G-mean
ReLU	0.9175	0.9104	0.9145	0.6016	0.9524
Sigmoid	0.9159	0.907	0.8783	0.6001	0.9524
Tanh	0.9159	0.907	0.8783	0.6001	0.9524

of precision and recall over all three data set are shown in Figures 3.

Based on the analysis of experimental results, it can be concluded that as the α value gradually increases, the values of different evaluation metrics continue to increase. When α reaches a certain value, the changes in the values of different evaluation metrics stabilizes.

This paper examines the impact of activation functions in SAD (Self-Attention Distillation) and presents a comparative analysis of the evaluation metric outcomes of SAD on various datasets, specifically focusing on the F1-G measure. Based on the data shown in Table VIII, it is apparent that the Rectified Linear Unit (ReLU) function demonstrates superior performance in precision, F1-G measure, and G-mean across three assessment criteria. Tables IX show that the ReLU function excels in various evaluation metrics.

According to the algorithm procedure in Section IV, the time complexity of SAD is $O(m) + O(\text{epoch} * 2m)$, which is $O(n^2)$, making it have a higher time complexity compared

to classical anomaly detection methods but similar to other baseline methods. With the same time complexity, this method shows more outstanding performance on different datasets.

VI. CONCLUSION

E-commerce networks involve relationships between different entities and have a structure of a heterogeneous information network. By analyzing anomalous user behavior patterns, a self-supervised anomaly detection method, SAD, is proposed for e-commerce networks. A self-supervised anomaly detection technique called SAD is presented for e-commerce networks. It operates by examining unusual user Behavior patterns. Within this theoretical framework, a method for identifying and detecting users who display aberrant Behavior has been proposed: the Self-Supervised Aberrant Detection Model (SAD). Experimental verification and comparative analysis demonstrate that SAD outperforms baseline methods in various evaluation metrics, proving the feasibility and superiority of this method. Despite having great anomaly detection performance, the SAD approach described in this paper ignores node and edge attribute information in heterogeneous information networks. Furthermore, although there are other links among various entities in e-commerce networks, this research primarily focuses on the relationships between users and devices. In the future, we hope to further expand research on anomaly user detection by analyzing behavior patterns among different entities.

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